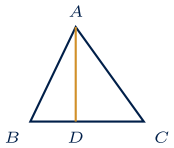
Subject: Mathematics | **PERIODIC COMPREHENSIVE TEST – 08** | **Time:** 60 Minutes
Class: Class X (CBSE) | **Chapter 6: Triangles** | **MM: 30**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Marks are indicated against each question.
2. **Section A** comprises 6 questions of 1 mark each (5 MCQs and 1 Assertion-Reason).
3. **Section B** comprises 2 Very Short Answer (VSA) questions of 2 marks each.
4. **Section C** comprises 2 Short Answer (SA) questions of 3 marks each.
5. **Section D** comprises 2 Long Answer (LA) questions of 5 marks each.
6. **Section E** comprises 1 Case Study based integrated unit of assessment of 4 marks.
7. Use of calculators is strictly prohibited. Draw neat, labeled geometric diagrams wherever required.

SECTION A: OBJECTIVE QUESTIONS (6 Marks)



1. In ΔABC, D is a point on side BC such that $\angle ADC = \angle BAC$. If $CA = 12$ cm and $CB = 16$ cm, then CD is:

(a) 8 cm (b) 9 cm (c) 10 cm (d) 6 cm

2. If $\Delta ABC \sim \Delta PQR$ with perimeters 35 cm and 45 cm respectively, and $PQ = 9$ cm, then AB is:

(a) 7 cm (b) 8 cm (c) 6 cm (d) 7.5 cm

3. In ΔABC, $DE \parallel BC$. If $AD = 4$ cm, $AE = 8$ cm, $DB = x - 4$, and $EC = 3x - 19$, then x equals:

(a) 9 (b) 10 (c) 11 (d) 12

4. Two vertical poles of height 6 m and 11 m stand on level ground. If their feet are 12 m apart, distance between their tops is:

(a) 12 m (b) 13 m (c) 14 m (d) 15 m

5. In ΔABC, AD is the bisector of $\angle A$ meeting BC at D. If $AB = 6$ cm, $AC = 4$ cm, and $BD = 3$ cm, then CD is:

(a) 2 cm (b) 2.5 cm (c) 3 cm (d) 1.5 cm

6. **Assertion (A):** Two similar triangles are always congruent.
Reason (R): Two congruent triangles are always similar.

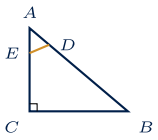
(a) Both (A) and (R) are true and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
(c) Assertion (A) is true but Reason (R) is false.
(d) Assertion (A) is false but Reason (R) is true.

SECTION B: VERY SHORT ANSWER QUESTIONS (4 Marks)

7. **[2 Marks]** A vertical stick of length 1.8 m casts a shadow 1.2 m long on level ground. At the same time, a vertical building casts a shadow 24 m long. Find the height of the building.

8. **[2 Marks]** In ΔABC, D and E are points on sides AB and AC respectively such that $\angle B = \angle C$ and $BD = CE$. Prove mathematically that $DE \parallel BC$.

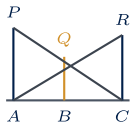
SECTION C: SHORT ANSWER QUESTIONS (6 Marks)



9. **[3 Marks]** In the given figure, ABC is a triangle right-angled at C and $DE \perp AB$. Prove that $\Delta ABC \sim \Delta ADE$ and find the lengths of AE and DE if $AB = 15$ cm, $AC = 12$ cm, and $AD = 3$ cm.

10. **[3 Marks]** In ΔABC, AD is an altitude to BC. If $BD = 8$ cm, $DC = 2$ cm, and $AD = 4$ cm, prove that ΔABC is right-angled at A.

SECTION D: LONG ANSWER / HOTS QUESTIONS (10 Marks)

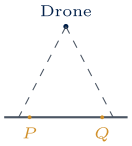


11. **[5 Marks]** In the given figure, PA, QB, RC are each perpendicular to line segment AC. If $PA = x, QB = y, RC = z, AB = a, BC = b$, prove that: (i) $\frac{y}{x} = \frac{b}{a+b}$, (ii) $\frac{y}{z} = \frac{a}{a+b}$, (iii) $\frac{1}{x} + \frac{1}{z} = \frac{1}{y}$.

12. **[5 Marks]** In a right triangle ABC right-angled at B, $BD \perp AC$. Prove that $\Delta ADB \sim \Delta ABC \sim \Delta BDC$. Hence deduce that: (i) $BD^2 = AD \cdot CD$, (ii) $AB^2 = AD \cdot AC$, and (iii) if $BD = p, AB = c, BC = a$, prove that $\frac{1}{p^2} = \frac{1}{a^2} + \frac{1}{c^2}$.

SECTION E: COMPETENCY / CASE STUDY QUESTION (4 Marks)

13. **Case Study 08: Drone Aerial Surveying & Ground Distance Resolution**



A surveying drone hovers at a fixed altitude of 120 m directly above a reference landmark. Its wide-angle downward camera captures a high-resolution image where two survey pins on level ground, P and Q, appear separated by a sensor focal distance of 6 cm. The camera's focal length is 50 mm (5 cm).

(i) **[1 Mark]** State the linear scale factor between the camera sensor image and the ground distances.

(ii) **[1 Mark]** What is the actual horizontal ground distance between survey pins P and Q?

(iii) **[2 Marks]** If another pair of ground markers is known to be 180 m apart, what distance will separate their images on the camera sensor?

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